Unit 2: Project Management Processes



Project Management Processes

Project management is the **application of knowledge, skills, tools, and techniques to project activities** to meet the project requirements and it is accomplished through the appropriate application and integration of the logically grouped project management processes, which are categorized into **five Process Groups.**

Project Management Processes

This module **discusses these process groups**. It highlights the key purposes and provides the detailed processes involved in each process group.

At the end, it illustrates the **relationship of each process group with the different knowledge areas** which will be taken up in separate modules in this instructional material.

Learning Outcomes

Identify and **define the different process groups**

Determine in what **process group** a specific **process/activity is mapped to**

Differentiate a **product life cycle** from a project life cycle

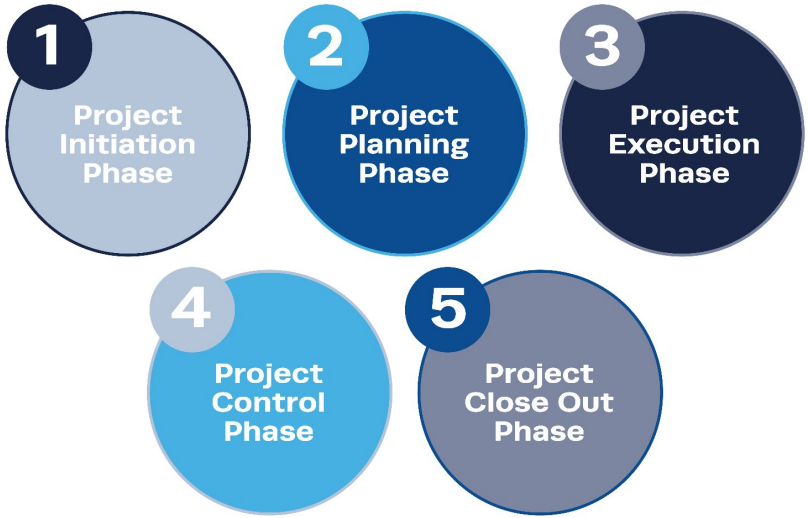
Explain the **key purposes** of each process group

Apply the **processes into real-life projects**



What Is Project Life Cycle?

Project Life Cycle is the series of phases that a project passes through from its initiation to its closure. All projects can be mapped to the following **generic life cycle structure.**



The project life cycle **helps ensure that projects are executed** in an organized, predictable, and efficient manner, **increasing the likelihood of success and reducing risks.**

Characteristics of a Generic Life Cycle

Cost and staffing levels are low at the start, peak as the work is carried out, and drop rapidly as the project draws to a close. This may not apply though to all projects.

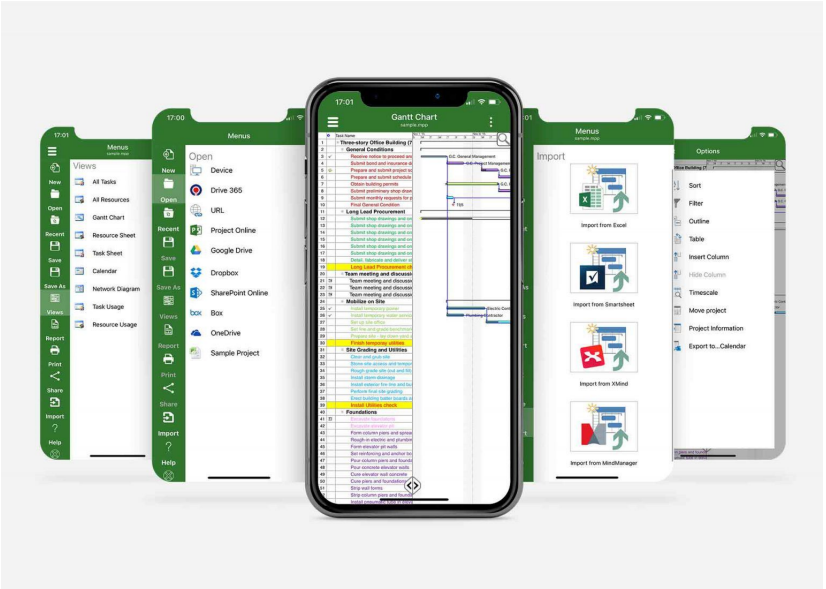
Risk and uncertainty are greatest at the start of the project. These factors decrease over the life of the project as decisions are reached and as deliverables are accepted.

The **ability to influence the final characteristics of the project's product**, without significantly impacting cost, is highest at the start of the project and decreases as the project progresses towards completion.

Characteristics of a Generic Life Cycle

Product Life Cycle - are the different stages in the life of a product, which are conception, growth, maturity, decline, and withdrawal

During the course of its life, **a product may need or require many projects.** For example, a project during product conception might be to determine the customer's needs. During maturity, a project might be to analyze the competition.



Stage	Characteristics	Typical Projects / Activities
Conception / Development	**Apple researches new technologies**, customer preferences (camera, battery, screen, 5G etc.), designs prototypes, tests hardware/software compatibility.	**R&D projects** to build new features; usability testing; technical feasibility studies; design & prototyping; sourcing components; regulatory/quality compliance.
Introduction	When a new **iPhone model is launched.** Marketing hype, early adopters, high cost, lower volume ramp-up.	**Product launch campaigns**; partner/distributor training; initial supply chain scale-up; gathering feedback and fixing early bugs; promotions to reviewers/influencers.
Growth	**Sales climb, production scales, distribution broadens globally**. Apple adds more features, maybe release variants (different storage, sizes). Competitive pressure from Android phones pushes innovation.	**Scaling manufacturing; optimizing supply logistics;** software updates based on feedback; refining features; expanding into more markets; aggressive marketing; maybe lowering prices or introducing trade-in programs.
Maturity	The **iPhone becomes well established**; most demand comes from replacement/upgrades rather than new customers. Competition intense. Growth slows. Margins may compress.	Focus on **cost reduction; optimization of production**; incremental improvements (camera, battery life, software); marketing campaigns to retain customers; premium features / bundled services (AppleCare, services); possibly launch of high-end / specialized models.
Decline	Eventually**, older models lose appeal**; technology shifts (e.g. toward foldables, AR, wearables, etc.) might make older designs less competitive. Some features become obsolete. Sales of older generation drop.	**Phasing out old models**; optimizing support; discontinuing less profitable variants; preparing successors; maybe repositioning the product (e.g. lowering price or focusing on emerging markets where the product is still competitive).
Withdrawal / Discontinuation	**Apple discontinues a model** (or series) when it no longer makes sense to maintain it: support cost, component availability, declining demand.	**Product retirement planning;** customer communication (support, parts); recycling or refurbishing programmes; shifting resources to newer models; final financial accounting of the product line.





Process Groups

Process Groups – are the categories by which the project management processes are logically grouped.

Process Groups are the **structured categories that organize project management activities throughout a project's life.**

They provide a logical flow, beginning with initiating the project, then planning the details, followed by executing the work.

Along the way, **monitoring and controlling ensures progress stays on track, and finally, closing formally completes and evaluates the project.** Together, these groups serve as a framework to manage projects efficiently and consistently.



Why are process groups important?

Understanding process groups is important because they provide **a clear framework for managing projects in an organized and systematic way.**

They help project managers and teams know what to do at each stage, **ensuring that nothing critical is overlooked.**

By following these groups, **teams can better align** resources, track progress, manage risks, and deliver results that meet objectives.

Ultimately, they **improve efficiency, consistency, and the likelihood of project success.**

Five Process Groups



Five Process Groups

Initiating | **Planning** | **Executing** | **Monitoring and Controlling** | **Closing**

The initiating process group includes the activities performed to define a new project or a new phase of an existing project by obtaining authorization to start. These processes set the **vision of the project**—clarifying what needs to be accomplished.

Key Purpose

- Align stakeholders' expectations with the project's purpose
- Provide visibility on the scope and objectives
- Show how stakeholder participation ensures expectations are met

Processes Involved in Initiating

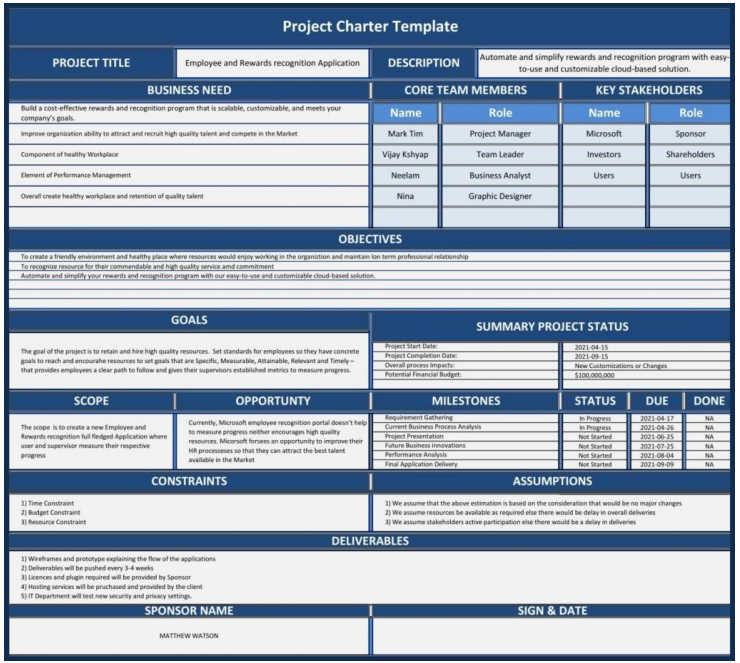
1. Identify the project manager
2. Identify stakeholders
3. Define initial scope
    - Uncover initial requirements and risks
    - Understand the business case
    - Review company culture and existing systems
    - Collect processes, procedures, and historical information
4. Break large projects into phases
5. Create measurable objectives
6. Commit initial financial resources
7. Develop the project charter

Five Process Groups

Initiating | **Planning** | **Executing** | **Monitoring and Controlling** | **Closing**

Project Charter Includes

- Project purpose or justification
- Measurable objectives and success criteria
- High-level requirements
- Assumptions and constraints
- High-level project description and boundaries
- High-level risks
- Summary milestone schedule
- Summary budget
- Stakeholder list
- Project approval requirements
- Assigned project manager, responsibilities, and authority
- Name and authority of the project sponsor



Project Charter Template

PROJECT TITLE	Employee and Rewards recognition Application	DESCRIPTION	Automate and simplify rewards and recognition program with easy-to-use and customizable cloud-based solution.

BUSINESS NEED	CORE TEAM MEMBERS		KEY STAKEHOLDERS	
	Name	Role	Name	Role
Build a cost-effective rewards and recognition program that is scalable, customizable, and meets your company's goals.	Mark Tim	Project Manager	Microsoft	Sponsor
Improve organization ability to attract and recruit high quality talent and compete in the Market	Vijay Kshyap	Team Leader	Investors	Shareholders
Component of healthy Workplace	Neelam	Business Analyst	Users	Users
Element of Performance Management	Nina	Graphic Designer		
Overall create healthy workplace and retention of quality talent				

OBJECTIVES

To create a friendly environment and healthy place where resources would enjoy working in the organization and maintain lon term professional relationship
To recognize resource for their commendable and high quality service amd commitment
Automate and simplify your rewards and recognition program with our easy-to-use and customizable cloud-based solution.

GOALS

The goal of the project is to retain and hire high quality resources. Set standards for employees so they have concrete goals to reach and encourahe resources to set goals that are Specific, Measurable, Attainable, Relevant and Timely – that provides employees a clear path to follow and gives their supervisors established metrics to measure progress.

SUMMARY PROJECT STATUS

Project Start Date:	2021-04-15
Project Completion Date:	2021-09-15
Overall process Impacts:	New Customizations or Changes
Potential Financial Budget:	$100,000,000

SCOPE	OPPORTUNTY	MILESTONES	STATUS	DUE	DONE
The scope is to create a new Employee and Rewards recognition full fledged Application where user and supervisor measure their respective progress	Currently, Microsoft employee recognition portal doesn't help to measure progress neither encourages high quality resources. Microsoft forsees an opportunity to improve their HR processeses so that they can attract the best talent available in the Market	Requirement Gathering	In Progress	2021-04-17	NA
		Current Business Process Analysis	In Progress	2021-04-26	NA
		Project Presentation	Not Started	2021-06-25	NA
		Future Business innovations	Not Started	2021-07-25	NA
		Performance Analysis	Not Started	2021-08-04	NA
		Final Application Delivery	Not Started	2021-09-09	NA

CONSTRAINTS

1) Time Constraint
2) Budget Constraint
3) Resource Constraint

ASSUMPTIONS

1) We assume that the above estimation is based on the consideration that would be no major changes
2) We assume resources be available as required else there would be delay in overall deliveries
3) We assume stakeholders active participation else there would be a delay in deliveries

DELIVERABLES

1) Wireframes and prototype explaining the flow of the applications
2) Deliverables will be pushed every 3-4 weeks
3) Licences and plugin required will be provided by Sponsor
4) Hosting services will be pruchased and provided by the client
5) IT Department will test new security and privacy settings.

SPONSOR NAME	SIGN & DATE
MATTHEW WATSON	

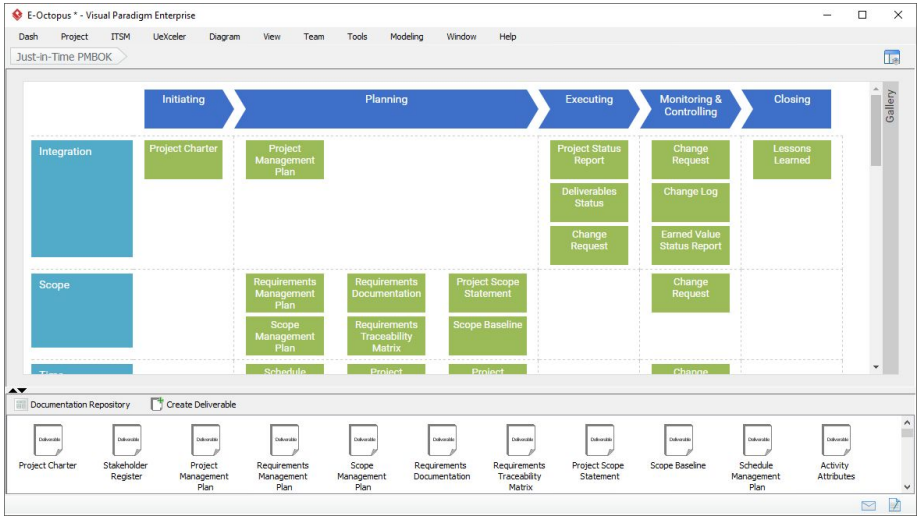
Five Process Groups

Initiating | **Planning** | Executing | Monitoring and Controlling | Closing

The planning process group consists of the activities needed to establish the project's scope, refine its objectives, and define the course of action required to achieve those objectives.

This stage ensures the project is well-structured and has a clear roadmap before execution begins.

Key Purpose
Define the strategy, tactics, and course of action to successfully complete the project or phase.

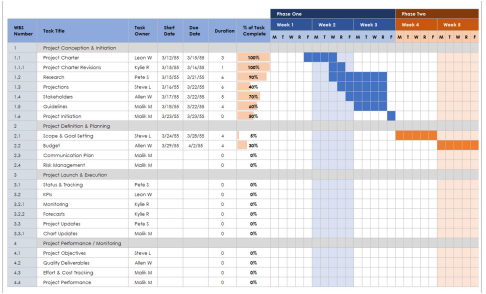


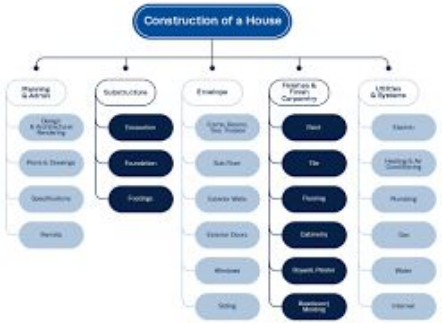
Five Process Groups

Initiating | **Planning** | Executing | Monitoring and Controlling | Closing

Processes Involved in Planning

1. Establish the total scope of effort
2. Finalize requirements
3. Define and refine objectives
4. Develop the course of action to achieve objectives:
    - Create activity list
    - Develop the Work Breakdown Structure (WBS)
    - Estimate resource requirements (identify team, roles, and responsibilities)
    - Estimate time and develop the project schedule (determine critical path, create network diagram)
    - Identify procurement needs
    - Estimate and develop budget (costs)
    - Define quality standards, processes, and metrics
    - Plan communications
    - Perform risk identification, qualitative and quantitative analysis, and risk response planning
    - Iterate plans as needed for refinement
5. Finalize execution and control components of all management plans
6. Develop the final project management plan and performance measurement baseline
7. Obtain formal approval of the plan
8. Conduct the project kickoff meeting





Five Process Groups



The executing process group involves carrying out the work defined in the project management plan to deliver the project's outputs and meet specifications.

It focuses on coordinating people, resources, and processes to ensure that the project delivers what was planned.

Key Purpose
Complete the work as defined in the project management plan to meet the project's specifications.

Processes Involved in Executing

1. **Integrate and perform project activities** according to the project management plan:
    - Produce the product scope
    - Request and implement approved changes
    - Ensure a common understanding among stakeholders
    - Continuously improve processes and outputs
    - Perform quality assurance and quality audits
2. **Coordinate people and resources, and manage stakeholder expectations:**
    - Acquire the final project team
    - Manage people effectively
        - Evaluate team and project performance
        - Conduct team-building activities
        - Provide recognition and rewards
        - Facilitate conflict resolution
    - Hold regular project meetings to ensure alignment

Five Process Groups



The **monitoring and controlling process group** involves tracking, reviewing, and regulating the progress and performance of the project.

It ensures the project stays **aligned with the plan and identifies areas requiring corrective action.** Continuous monitoring provides insight into the project's health and helps manage deviations before they escalate.

Key Purpose

- Measure and analyze project performance at regular intervals or key events to identify variances from the project management plan.
- Ensure the entire project effort is monitored and controlled for alignment with objectives.

Processes Involved in Monitoring and Controlling

1. **Track, review, and manage project progress and performance:**
   - Monitor ongoing activities against the project management plan and performance measurement baseline
   - Identify variances and determine if a change request is needed
2. **Identify and implement necessary changes:**
   - Recommend corrective or preventive actions to address potential problems
   - Approve or reject changes and perform change control
   - Inform stakeholders of approved changes
3. **Perform quality control** to ensure outputs meet standards
4. **Conduct risk audits** to assess and manage project risks

Five Process Groups

Initiating → Planning → Executing → Monitoring and Controlling → **Closing**

The closing process group involves **finalizing all activities** across all process groups to formally close a project or phase.

This ensures that the project is **completed according to plan, all objectives are met, and any premature closure** (e.g., cancelled or aborted projects) is properly documented.

Key Purpose

- Verify that all processes are completed within each process group to formally close the project or phase.
- Establish proper closure procedures for projects that end prematurely.

Processes Involved in Closing

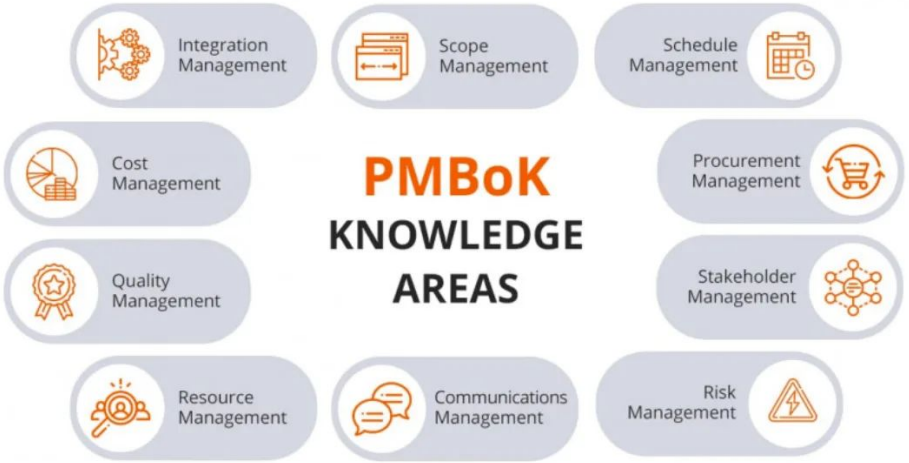
1. **Confirm work is completed to requirements:**
   - Gain formal acceptance of the product or deliverables
2. **Update organizational process assets:**
   - Complete final performance reporting
   - Update the knowledge base
   - Index and archive project records
3. **Close procurement activities:**
   - Terminate all relevant agreements and contracts
4. **Hand off completed product** to the customer or operational team
5. **Assess and release project team members:**
   - Conduct performance evaluations
   - Release resources back to the organization

Role of the Knowledge Areas

The project management processes are **further grouped into ten separate Knowledge Areas.**

A **Knowledge Area** represents a complete set of concepts, terms, and activities that make up a professional field, project management field, or area of specialization. These ten Knowledge Areas are used on most projects most of the time.

Project teams should utilize these ten Knowledge Areas and other Knowledge Areas, **as appropriate, for their specific project.**



Mapping the processes in each Process Group with the Knowledge Areas

Mapping the processes in each **Process Group** with the **Knowledge Areas** is important because it provides a **clear framework for managing all aspects of a project**.

It shows how the activities of initiating, planning, executing, monitoring & controlling, and closing relate to key project knowledge domains like scope, time, cost, quality, risk, communications, procurement, and stakeholders.

ℹ️ *In short, it acts as a roadmap for project managers to integrate processes and knowledge areas effectively, ensuring nothing is overlooked and the project stays aligned with its objectives.*

Ensure completeness: It guarantees that every aspect of project management is addressed at each stage of the project.

Provide structure: Helps project managers see which processes apply at which stage and in which knowledge area.

Improve coordination: Facilitates better planning, execution, and control by linking activities to their relevant knowledge areas.

Support decision-making and risk management: Enables systematic tracking of project performance, resource use, and risk mitigation.

Mapping the processes in each Process Group with the Knowledge Areas

PROCESS GROUP AND KNOWLEDGE AREA MAPPING

Knowledge Areas	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	
9. Project Resource Management		9.1 Plan Resource Management 9.2 Estimate Activity Resources	9.3 Acquire Resources 9.4 Develop Team 9.5 Manage Team	9.6 Control Resources	
10. Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	
11. Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
12. Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
13. Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder Engagement	

Developed by: DEEP FRIED BRAIN PROJECT

Source: *PMBOK® Guide*, 6th Edition

Case Study: IKEA – Development of the IKEA Place AR Application



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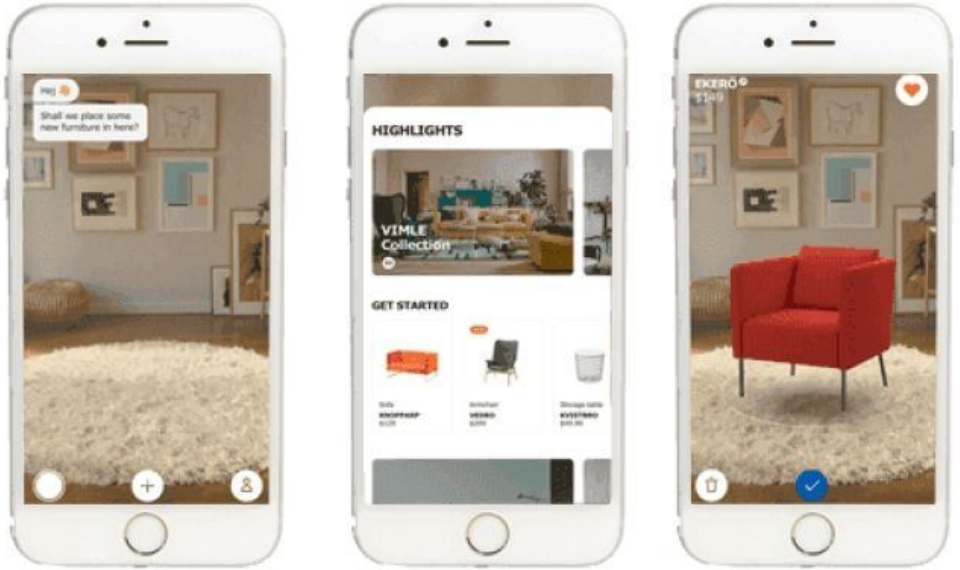
Company Overview:

IKEA is a global leader in home furnishings, renowned for its innovative approach to retail and commitment to enhancing customer experiences.

Project Overview:

In September 2017, IKEA launched the **IKEA Place app,** an **augmented reality application** designed to allow customers to **visualize how IKEA's furniture would look** in their own homes before making a purchase.

The app utilizes **Apple's ARKit technology** to provide a realistic and interactive shopping experience.



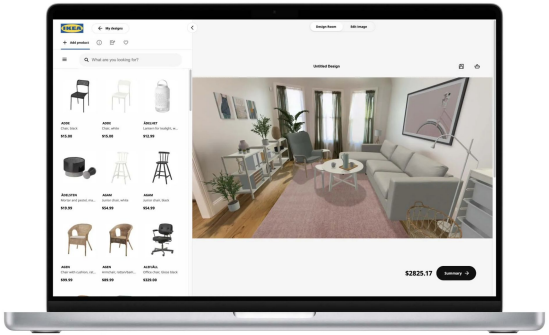
Case Study: IKEA – Development of the IKEA Place AR Application

Initiating | **Planning** | **Executing** | **Monitoring and Controlling** | **Closing**

- **Project Charter:** The project was initiated to address the challenges customers face in visualizing how furniture would fit and look in their living spaces, aiming to enhance the online shopping experience.

- **Stakeholder Identification:** Key stakeholders included IKEA's digital innovation team, product designers, software developers, marketing department, and end-users.

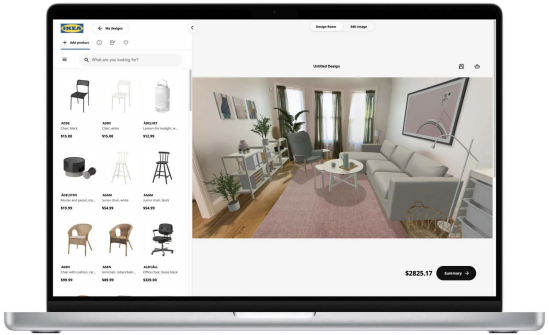
- **Initial Scope:** The app was designed to support a wide range of IKEA's product catalog, allowing users to place true-to-scale 3D models of furniture in their real-world environments.



Case Study: IKEA – Development of the IKEA Place AR Application

Initiating | **Planning** | Executing | Monitoring and Controlling | Closing

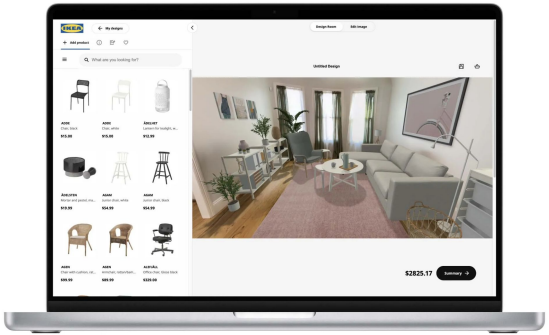
- **Scope Definition:** Detailed planning sessions were held to define the app's features, user interface, and integration with IKEA's existing e-commerce platform.
- **Schedule Development:** A timeline was established, with milestones for app development, testing, and launch.
- **Budgeting:** A financial plan was created, estimating costs for development, marketing, and maintenance.
- **Risk Management:** Potential risks such as technological challenges, user adoption rates, and integration issues were identified, with mitigation strategies developed.



Case Study: IKEA – Development of the IKEA Place AR Application

Initiating | Planning | **Executing** | Monitoring and Controlling | Closing

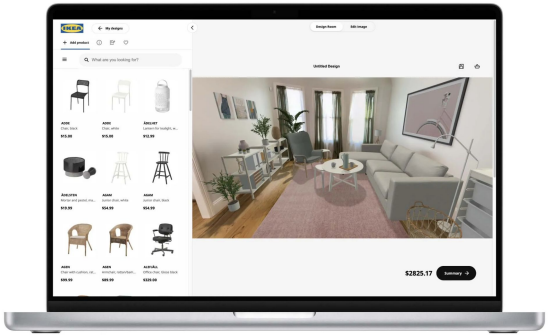
- **Team Acquisition:** A project team was assembled, including software developers, UI/UX designers, and marketing personnel.
- **Quality Assurance:** Regular testing and feedback loops were implemented to ensure the app met quality standards and user expectations.
- **Stakeholder Engagement:** Continuous communication with stakeholders kept them informed and involved, addressing concerns promptly.
- **Procurement Management:** Technology partners were selected to provide the necessary AR development tools and platforms.



Case Study: IKEA – Development of the IKEA Place AR Application

Initiating | Planning | Executing | **Monitoring and Controlling** | Closing

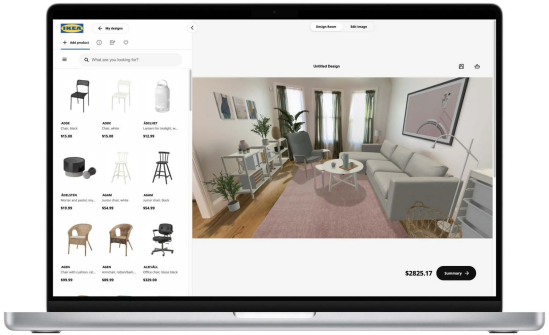
- **Performance Tracking:** Key performance indicators (KPIs) such as app downloads, user engagement, and sales conversions were regularly monitored.
- **Change Management:** Any deviations from the plan were addressed through formal change requests, with impacts assessed and approvals obtained.
- **Risk Monitoring:** Identified risks were continuously assessed, and mitigation plans were adjusted as necessary.
- **Reporting:** Regular reports were generated for stakeholders, providing updates on project status and any issues encountered.



Case Study: IKEA – Development of the IKEA Place AR Application

Initiating | Planning | Executing | Monitoring and Controlling | **Closing**

- **Final Acceptance:** Upon completion, the app underwent a final review, and any necessary adjustments were made.
- **Handover:** The app was officially launched and made available to customers.
- **Post-Implementation Review:** A review was conducted to assess project outcomes against objectives, identifying lessons learned for future projects.



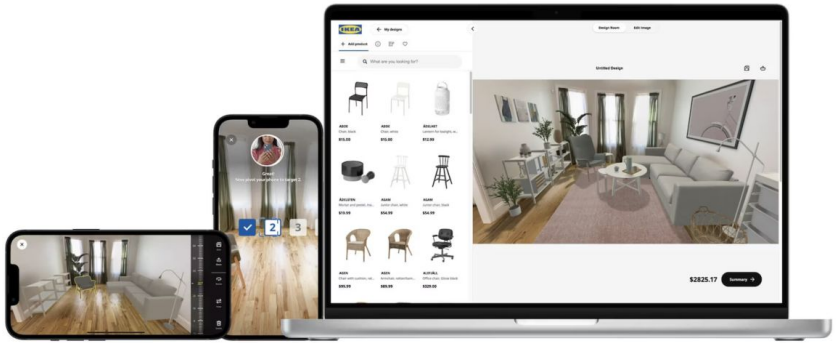
Case Study: IKEA – Development of the IKEA Place AR Application

The **result of the project management cycle** for IKEA's **IKEA Place AR application** was the successful development and launch of an augmented reality app that allows customers to visualize furniture in their own homes before purchasing. By following all five **process groups**—initiating, planning, executing, monitoring & controlling, and closing—**IKEA ensured the project stayed on track, met quality standards, stayed within budget, and aligned with stakeholder expectations**.

How it helped IKEA:

1. **Enhanced Customer Experience:** Customers could virtually place IKEA furniture in their spaces, reducing uncertainty and improving decision-making.
2. **Increased Sales and Engagement:** The app helped drive online engagement and boosted confidence in purchasing, likely increasing conversion rates.
3. **Competitive Advantage:** Leveraging AR technology positioned IKEA as an innovative leader in retail, differentiating it from competitors.
4. **Efficient Resource Management:** The structured project management approach allowed IKEA to coordinate teams, manage risks, and allocate resources effectively.
5. **Data for Future Projects:** Insights from app usage and customer feedback informed improvements in product offerings, marketing, and technology adoption.

In short, the project management cycle turned an innovative idea into a **tangible, functional, and strategic tool** that directly benefited IKEA's **business operations and customer satisfaction.**



Assessment

Objective: This activity helps students understand how process groups and knowledge areas function in project management. It connects theoretical concepts to real-world projects, fostering critical thinking and practical application of project management principles.

- **Choose a Knowledge Area** – e.g., Scope, Time, Cost, Quality, Risk, Communications, Procurement, or Stakeholder Management.
- **Select a Brand and a Real Project** – It could be a product launch, marketing campaign, new store opening, software development, or any identifiable project by a brand.
- **Analyze the Project Using Process Groups:**
  - **Initiating:** How was the project started? Who were the stakeholders? Was a charter created?
  - **Planning:** How were objectives defined? How was scope, schedule, and budget planned?
  - **Executing:** How were resources coordinated? How was the plan implemented?
  - **Monitoring & Controlling:** How was progress tracked? Were risks and changes managed?
  - **Closing:** How was the project finalized? Were lessons learned documented?

Output: Presentation showing how the project applied each process group in the chosen knowledge area.

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